Describe transformations of parent functions from the equation

Pre-Cal | CK-12: Graphical Transformations

Key Idea

Transformations move or reshape a parent graph. In $y = a\,f(x-h)+k$: h shifts left/right, k shifts up/down, a stretches (and flips if negative).

Key Terms

- **Translation** — a slide (from h or k)
- **Reflection** — a flip (from a negative a)

Worked steps

Step 1: Match the equation to $a\,f(x-h)+k$.
Step 2: Read h (opposite sign), k, and a.
Example: $y = -(x-2)^2+5$ — right 2, up 5, and flipped over the x-axis.

You Try

Describe $y = (x+3)^2 - 1$ from the parent $y = x^2$.